A Real-time Research Project Report

on

INTEGRITYX

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

by

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Department of Computer Science & Engineering

BVRIT HYDERABAD College of Engineering for Women

(NAAC Accredited-A Grade | NBA Accredited B.Tech. (EEE, ECE & CSE)

(UGC Autonomous,Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad)

Bachupally, Hyderabad – 500090

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CERTIFICATE

This is to certify that the Real-time Research Project report on "**INTEGRITYX**" is a bonafide work carried out by **Ms. G. PRANATHI (24WH1A0584), Ms. R. PUJITHA (24WH1A0597), and Ms. M. PRASANNA (24WH1A05A5)** in the partial fulfillment for the award of B.Tech. degree in **Computer Science & Engineering**, **BVRIT HYDERABAD College of Engineering for Women**, Bachupally, Hyderabad, affiliated to Jawaharlal Nehru Technological University Hyderabad, under my guidance and supervision. The results embodied in the real-time research project work have not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the work presented in this Real-time Research Project entitled **"INTEGRITYX"** submitted towards completion of Real-time research project work in II B.Tech., CSE at 'BVRIT HYDERABAD College of Engineering for Women', Hyderabad is an authentic record of our original work carried out under the guidance of **Mr. M.N. Sriharsha,** Assistant Professor, Department of CSE.

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ABSTRACT

IntegrityX is a GitHub-based automated assignment evaluation system designed to simplify and improve the evaluation of programming assignments in academic institutions. Traditional evaluation methods require instructors to manually verify submissions, execute programs, and assign marks, which is often time-consuming and prone to inconsistencies. IntegrityX addresses these challenges by introducing an automated evaluation workflow that integrates GitHub repositories, backend processing, and test case execution. In this system, students submit assignments by pushing their code to GitHub repositories, where each repository serves as a version-controlled submission space. Whenever a student pushes code, a GitHub webhook is triggered automatically, notifying the backend server about the new submission and acting as the main event trigger for evaluation. Upon activation, the backend system retrieves the submitted code and executes it against predefined instructor- defined test cases. The evaluation follows a Level 1 assessment approach, where the final score is calculated based on the number of passed test cases, ensuring uniform grading criteria and reducing human bias. A major advantage of IntegrityX is its ability to provide real-time feedback, enabling students to immediately view results after each code push and improve their solutions efficiently. This encourages continuous learning and promotes modern software development practices such as iterative development and automated testing. Overall, IntegrityX reduces the manual workload for instructors, ensures fair and consistent evaluation, and aligns academic assessment methods with real-world software engineering practices using tools like GitHub and automated evaluation pipelines.

Keywords: *Automated Evaluation, GitHub Integration, Programming Assignments, Web-hooks, Test Case Execution, Real-time Feedback, Continuous Assessment, Academic*

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LIST OF TERMS AND ABBREVIATIONS

JWT	JSON Web Token
RBAC	Role Based Access Control
MERN	MongoDB, Express.js, React.js, Node.js
URL	Uniform Resource Locator
CRUD	Create, Read, Update, Delete
REST	Representational State Transfer
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 Introduction

In recent years, the rapid growth of software development practices and digital learning environments has significantly transformed the way programming education is delivered and evaluated. Programming assignments play a crucial role in engineering education, particularly in disciplines related to computer science and information technology. These assignments help students develop problem-solving abilities, logical thinking, and practical coding skills. However, despite technological advancements, the evaluation of programming assignments in many academic institutions continues to rely on traditional manual methods. Faculty members are often required to collect submissions, execute code individually, verify outputs, and assign grades. This process is not only time-consuming but also prone to inconsistencies, especially when handling large numbers of students.

At the same time, modern software development practices in the industry heavily depend on version control systems, with GitHub being one of the most widely used platforms. GitHub enables developers to manage code efficiently, track changes, collaborate with others, and maintain version history. Despite its importance in real-world software development, GitHub is rarely integrated into academic assignment evaluation systems. Most existing academic evaluation approaches focus on static submission formats such as file uploads, which do not reflect industry practices. Additionally, widely used online coding platforms are primarily designed for competitive programming and recruitment assessments, rather than structured academic evaluation.

To address these challenges, there is a growing need for a system that bridges the gap between academic evaluation methods and industry practices. IntegrityX is designed as a GitHub-based automated assignment submission and evaluation platform that modernizes programming assessment in academic institutions. The system allows students to submit assignments through public GitHub repositories, thereby encouraging the use of version control and improving familiarity with professional development tools.

The core functionality of IntegrityX lies in its ability to automatically fetch code from student repositories using GitHub APIs and evaluate it based on predefined criteria. These criteria may include test case execution, output verification, and adherence to assignment

requirements. Once the evaluation is complete, the system records the results in a structured database and updates a leaderboard that reflects student performance. This automated approach significantly reduces the workload on faculty members, while ensuring consistency and fairness in grading.

Another important aspect of IntegrityX is its focus on transparency and scalability. Students can clearly understand how their submissions are evaluated, as the system provides structured feedback based on test case results. Faculty members can efficiently manage assignments, define evaluation parameters, and monitor student performance through a centralized interface. The platform is designed to handle large numbers of submissions, making it suitable for institutions with high student intake.

Furthermore, the use of GitHub repositories introduces several advantages, such as maintaining version history, enabling incremental development, and promoting better coding practices. Students can continuously improve their submissions while maintaining a record of their progress. This not only enhances learning outcomes but also aligns academic practices with industry standards. By integrating automated evaluation with real-world development tools, IntegrityX creates a more practical and engaging learning environment.

In addition to improving efficiency and transparency, IntegrityX also addresses the issue of scalability in programming assessments. Manual evaluation becomes increasingly difficult as class sizes grow, leading to delays and potential inconsistencies in grading. Automated evaluation ensures that all submissions are processed uniformly and within a shorter time frame. The leaderboard feature further motivates students by providing a competitive yet structured view of performance, encouraging continuous improvement.

Overall, IntegrityX represents a significant step toward modernizing academic evaluation systems for programming assignments. By leveraging GitHub integration, automated evaluation techniques, and structured performance tracking, the platform offers a comprehensive solution that benefits both students and faculty. It not only reduces manual effort but also promotes fairness, transparency, and alignment with industry practices. As educational institutions continue to adopt digital solutions, systems like IntegrityX have the potential to redefine how programming skills are assessed and developed in academic environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Review of literature

The development of digital learning platforms has significantly transformed the way academic institutions manage assignments, evaluations, and student performance tracking. Traditional methods such as paper-based submissions, email attachments, and manual grading have long been associated with challenges including delayed evaluation, lack of transparency, and increased workload for faculty members. With the advancement of web technologies, academic systems have evolved toward structured online platforms that simplify assignment creation, submission, and assessment workflows.

Existing web-based assignment management systems have demonstrated that structured digital portals improve submission consistency and reduce administrative complexity. When assignments are presented with clear titles, descriptions, deadlines, and evaluation criteria, students are better able to understand expectations and complete their work efficiently. These systems also provide dedicated interfaces for both students and faculty, improving usability and reducing confusion in academic environments.

2.2 Web-Based Assignment Management Systems

Web-based assignment management systems have become an essential part of modern academic infrastructure by replacing outdated manual workflows with centralized digital platforms that allow faculty to create and distribute assignments efficiently while enabling students to submit their work in an organized manner. This transition from physical to digital systems has been widely recognized as a necessary step toward improving academic efficiency and reducing administrative burden, particularly as institutions continue to grow in size and complexity. Research indicates that structured digital portals improve submission consistency and reduce administrative challenges by presenting assignments with clear titles, descriptions, and evaluation criteria, helping students better understand expectations and complete their work effectively [1]. Additionally, platforms that provide dedicated interfaces for both students and faculty reduce confusion and enhance usability, while class-based assignment distribution ensures that students access only relevant content, thereby improving focus and minimizing interface clutter. IntegrityX adopts these best practices by offering separate and clearly defined dashboards for faculty and students.

2.3 Role-Based Authentication and Access Control

Security and controlled access are critical components of academic platforms that handle sensitive student and faculty data, making Role-Based Access Control (RBAC) an essential security model for ensuring that users can access only the functionalities relevant to their roles. In academic environments, this distinction is particularly important as students and faculty have fundamentally different responsibilities, and unrestricted access could compromise both system integrity and evaluation fairness. Students are limited to viewing assignments, submitting their work, and accessing their own submission records, while faculty members are granted privileges to create and manage assignments, review submissions, and assign marks, ensuring a clear separation of responsibilities that enhances both security and usability. Research on academic evaluation systems indicates that structured and role-aware platforms improve workflow efficiency and reduce user confusion by providing focused and role-specific interfaces [1]. To implement this securely in modern web applications, JSON Web Tokens (JWT) are widely used as a stateless authentication mechanism, where a digitally signed token containing user identity and role information is verified with each request.

2.4 GitHub Repository-Based Submission Systems

GitHub repository-based submission systems have become increasingly important in programming-focused academic environments, offering a more advanced alternative to traditional file uploads by providing complete version history, commit tracking, and visibility into a student's development process. This approach enables faculty to evaluate not just the final output but also the methodology and progression of the work, while helping students build essential version control skills widely used in professional software development. Research has shown that students using GitHub for submissions develop a stronger understanding of version control and collaborative workflows, while faculty benefit from deeper insights through access to code structure and commit history, making repository-based submissions more informative than static files [2]. The study also emphasizes that integrating GitHub into academic workflows helps bridge the gap between theoretical learning and real-world practices [2]. IntegrityX incorporates these findings by using GitHub repository links as the standard submission method, allowing students to submit public repository URLs through the assignment dashboard and enabling faculty to access them instantly via direct links, ensuring a simple, transparent, and effective evaluation process aligned with modern software engineering practices.

2.5 Manual Evaluation and Marks Management

Manual evaluation remains a vital component of programming assessments despite the rise of automated grading tools, as factors such as code quality, logical structure, readability, and problem-solving approach cannot be fully judged by automation alone. Research highlights that integrated grading interfaces—where faculty can access submissions and

assign marks within a single unified panel—significantly enhance efficiency and consistency by eliminating the need to switch between multiple tools, which often slows down the evaluation process [1]. The study also emphasizes the importance of allowing mark revisions to support fair re-evaluation and the benefit of real-time visibility of new submissions to reduce delays and keep faculty updated [1]. IntegrityX adopts these principles by providing a streamlined faculty submissions panel that displays each student's email, assignment title, and a direct GitHub repository link in one place, enabling faculty to review code, assign marks, and save results seamlessly within the same interface, thereby ensuring a simple, efficient, and effective evaluation workflow.

2.6 Research Gap and Proposed Solution

A review of existing literature highlights a significant gap in academic programming assessment systems, as most tools focus on only one or two aspects of the workflow rather than offering a fully integrated solution. Online judge systems are effective for automated code execution and test-case-based evaluation but lack features such as structured assignment management, role-based dashboards, and faculty-oriented grading support [1]. In contrast, repository-based classroom tools provide strong support for assignment distribution and GitHub-based submissions, offering transparency and insight into student work, but often fall short in areas like marks management, structured result recording, and unified evaluation interfaces for faculty [2]. This disconnect creates a clear research gap where no single platform combines assignment management, secure role-based access, repository-based submissions, and comprehensive faculty evaluation. IntegrityX addresses this gap by integrating all these components into one cohesive system, drawing on the strengths identified in prior research to deliver a more complete, efficient, and practical solution for programming-based academic assessment.

CHAPTER 3

PROBLEM STATEMENT

3.1 Problem Statement

There is currently no unified academic platform that effectively integrates GitHub-based assignment submission with automated evaluation and institutional access control. In many educational institutions, programming assignments are still submitted through traditional methods such as file uploads or email, which do not align with modern software development practices. At the same time, GitHub has become an essential tool in the software industry for version control, collaboration, and project management. However, its potential remains largely underutilized in academic evaluation systems. The absence of a structured platform that connects GitHub repositories with institutional workflows creates a gap between academic learning and real-world development environments. As a result, students miss the opportunity to gain practical exposure to industry-standard tools while faculty members struggle to efficiently manage and evaluate submissions.

Manual grading of programming assignments further adds to the complexity, as it is time-consuming, difficult to scale, and often lacks transparency and consistency. Faculty members are required to individually execute code, verify outputs, and assign marks, which becomes increasingly challenging as the number of students grows. Existing online platforms primarily focus on competitive programming or recruitment-based assessments and do not support structured academic evaluation with features such as push-based automated execution, secure access control, and real-time performance tracking. These limitations highlight the need for a secure and scalable system that can automatically evaluate GitHub-based submissions, compute scores accurately using predefined criteria, and present results through a transparent leaderboard. Such a platform would not only reduce manual effort but also ensure fairness, consistency, and improved learning outcomes in programming education.

CHAPTER 4

PROPOSED METHODOLOGY

The system follows a GitHub-integrated automated evaluation approach designed to streamline the process of programming assignment assessment in academic environments. Traditional evaluation methods rely heavily on manual checking, where faculty members individually execute student programs, verify outputs, and assign marks. This approach becomes inefficient and inconsistent when dealing with a large number of submissions. IntegrityX addresses these challenges by introducing an automated workflow that integrates GitHub repositories with backend evaluation mechanisms. The system ensures that every submission is processed in a standardized manner, reducing human effort while improving accuracy and transparency. By leveraging GitHub as the primary submission platform, students are encouraged to adopt industry-level version control practices, while faculty members benefit from automated grading and real-time performance tracking. The overall objective of this system is to create a scalable and efficient environment for programming assignment evaluation that aligns academic practices with modern software development workflows.

4.1 Methodology

The methodology of the system is based on a step-by-step automated evaluation process that ensures smooth execution of assignment submission and grading. The first step involves students submitting their code through GitHub repositories assigned to them. Each student is provided with a unique repository where they can push their assignment solutions. This ensures that all submissions are organized, version-controlled, and easily traceable.

The second step involves the activation of a GitHub webhook. Whenever a student pushes new code or updates an existing repository, the webhook is automatically triggered. This event-based mechanism eliminates the need for manual submission notifications and ensures real-time communication between GitHub and the backend system.

In the third step, the backend system retrieves the latest version of the code from the student's GitHub repository using GitHub APIs. This ensures that only the most recent submission is considered for evaluation, preventing outdated or incomplete versions from being graded.

The fourth step involves the execution of predefined test cases provided by the lecturer. These test cases are designed to validate the correctness of the submitted program. The system runs the student's code in a controlled environment and supplies different input values to test its functionality under various conditions.

In the fifth step, the system evaluates the results obtained from executing the test cases. The output generated by the student's program is compared with the expected output defined in the test cases. Based on this comparison, the system calculates the number of test cases successfully passed by the student. This forms the basis for the final score.

The final step involves storing and displaying the computed results. All evaluation data, including student details, repository links, test case results, and final scores, are stored in a centralized database. These results are then displayed on the student and faculty dashboards for review. Additionally, the system updates performance metrics to reflect student progress and ranking in real time.

4.2 User Interface and Implementation Screens

The IntegrityX system provides a structured and user-friendly interface designed separately for students and faculty members. The interface ensures ease of navigation and simplifies interaction with the system's core functionalities. The implementation screens are divided into several modules, each serving a specific purpose in the overall workflow.

4.2.1 Student Login Page

The Student Login Page serves as the entry point for students to access the system. It includes authentication fields such as username and password. The login process ensures that only registered and authorized students can access their assignments and submissions. Security mechanisms are implemented to protect user credentials and prevent unauthorized access. Once login is successful, students are redirected to their personalized dashboard where they can view assignments and track their performance.

4.2.2 Faculty Login Page

The Faculty Login Page is designed for instructors to securely access the administrative functionalities of the system. Faculty members use this interface to create assignments, define test cases, and monitor student submissions. The authentication system ensures that only authorized faculty personnel can manage academic content. After successful login, faculty members are directed to their dashboard, where they can manage all assignment-related activities.

4.2.3 Student Registration Page

The Student Registration Page allows new users to create an account in the system. It collects essential information such as student name, roll number, email address, department, and

GitHub username. The GitHub username is particularly important as it links the student's account to their repository submissions. Once the registration process is completed, the system stores the information in the database and enables login access for the student.

4.2.4 Student Dashboard

The Student Dashboard acts as the central interface for students after login. It provides a comprehensive view of all assigned programming tasks, submission status, evaluation results, and performance scores. Students can track whether their submissions have been evaluated and view detailed results of each test case. The dashboard also displays leaderboard rankings, encouraging students to improve their performance. This module enhances user engagement by providing real-time feedback on submissions.

4.2.5 Faculty Dashboard – Assignment Creation

This section allows faculty members to create and manage programming assignments. Faculty can define assignment titles, descriptions, deadlines, difficulty levels, and evaluation criteria. They can also upload or define test cases that will be used for automated evaluation. This structured approach ensures that all assignments follow a consistent format and evaluation process. Once created, assignments are made visible to all enrolled students.

4.2.6 Faculty Dashboard – Assignment View

The Assignment View module provides faculty members with an organized list of all created assignments. It allows them to monitor active, pending, and completed assignments. Faculty members can also edit assignment details if necessary, such as updating deadlines or modifying instructions. This module ensures efficient management of academic tasks and provides a clear overview of all ongoing activities.

4.2.7 Faculty Dashboard – Submissions

The Submissions section displays all student submissions retrieved directly from GitHub repositories. Faculty members can view evaluation results, test case performance, and overall student progress. The system highlights the number of passed and failed test cases for each submission. This module significantly reduces manual grading effort by automatically presenting evaluated results. Faculty members can also use this data to analyze class performance and identify students who may need additional support.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 Overview

The system architecture of IntegrityX is designed to automate the evaluation of programming assignments by integrating GitHub-based submission workflows with a backend evaluation engine. Traditional academic evaluation methods require instructors to manually download student submissions, execute programs individually, verify outputs, and assign marks. This process becomes inefficient, time-consuming, and error-prone when handling large classes or multiple assignments.

IntegrityX addresses these limitations by introducing an automated, scalable, and structured evaluation pipeline. The architecture is built around real-time event-driven processing using GitHub webhooks. Whenever a student submits or updates code in their GitHub repository, the system automatically triggers the evaluation process without requiring any manual intervention from faculty members.

The primary objective of this architecture is to ensure automation of assignment evaluation, reduction of manual grading effort, standardized assessment criteria, real-time feedback generation, and improved transparency in grading. By leveraging GitHub as the submission platform, students also gain exposure to version control systems and professional development workflows.

5.2 System Architecture Diagram

5.2.1 GitHub Repository

The GitHub Repository serves as the primary submission platform for students. Each student is assigned a dedicated repository where programming assignments are uploaded and maintained using version control. This ensures organized storage, traceability, and timestamp-based submission tracking.

5.2.2 Webhook Trigger

The Webhook trigger acts as the communication bridge between GitHub and the backend

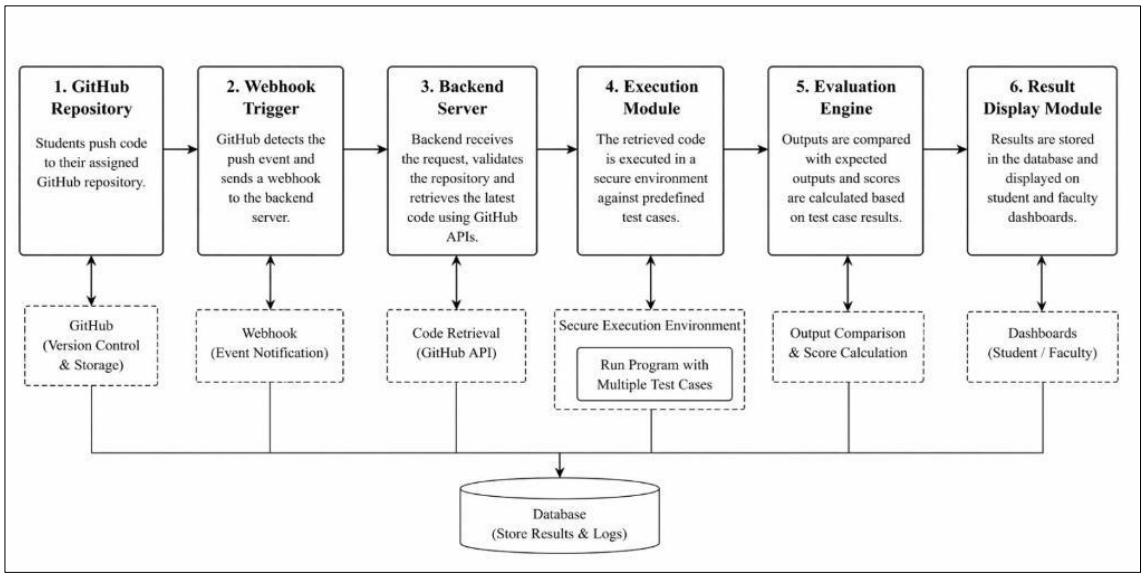


Fig 5.2.2.1: System Architecture of IntegrityX

server. Whenever students push new code or update existing files, GitHub automatically sends a webhook event to the backend, initiating the evaluation workflow in real time.

5.2.3 Backend Server

The backend server acts as the central controller of the entire system. It receives webhook requests, validates repository details, retrieves the latest code using GitHub APIs, coordinates test case execution, and stores evaluation results in the database.

5.2.4 Test Case Execution Module

This module is responsible for executing student programs against predefined test cases provided by faculty members. The code is executed in a secure and controlled environment to validate correctness and handle multiple input scenarios.

5.2.5 Evaluation Engine

The evaluation engine compares the actual output generated by the student's program with the expected output. Based on the number of passed test cases, it calculates scores and generates evaluation metrics.

5.2.6 Result Display Module

The result display module presents the evaluation outcomes through dashboards for both students and faculty members. It provides marks, passed test cases, failed cases, and submission history.

5.3 Workflow of the System

The workflow of IntegrityX follows an automated step-by-step evaluation pipeline.

1. **Student Pushes Code:** Students write their program locally and push it to the assigned GitHub repository.
2. **Webhook Trigger Activation:** GitHub automatically detects the push event and sends a webhook notification to the backend server.
3. **Code Retrieval:** The backend server retrieves the latest version of the submitted code using GitHub APIs.
4. **Execution of Test Cases:** The code is executed in a secure environment against multiple predefined test cases.
5. **Evaluation of Results:** The outputs are compared with expected outputs and marks are calculated.
6. **Display of Results:** Final scores and evaluation details are stored in the database and displayed on dashboards.

CHAPTER 6

EXPECTED OUTCOMES

6.1 Overview

The proposed IntegrityX system is expected to transform the conventional method of programming assignment evaluation by introducing an automated, scalable, and transparent assessment environment. The system integrates GitHub repositories with backend evaluation mechanisms to streamline submission tracking and result management. It minimizes manual intervention from faculty members and ensures that all students are evaluated under uniform conditions. By automating major parts of the evaluation workflow, the system improves efficiency, reduces errors, and supports fair grading practices. The expected outcomes include improvements in academic assessment quality, student engagement, and future expansion capabilities.

6.2 Current Outcomes

6.2.1 Improved Efficiency in Assignment Evaluation

The system is expected to significantly reduce the time and effort required for evaluating programming assignments. In traditional methods, faculty members manually execute student programs and verify outputs, which becomes difficult for large classes. With IntegrityX, the submission management and result tracking processes are streamlined through automation support. This improves the speed of academic evaluation and allows faculty to focus more on teaching and mentoring. The system thus increases operational efficiency and supports timely academic workflows.

6.2.2 Transparent and Fair Assessment Process

One of the key expected outcomes is the establishment of a transparent and unbiased evaluation process. Since all submissions are managed through a standardized workflow, every student's assignment is processed using the same criteria and guidelines. This eliminates the possibility of inconsistency or human bias in grading. Students can clearly understand how their assignments are evaluated, which builds trust in the system. Such transparency enhances the credibility and fairness of the academic assessment process.

6.2.3 Centralized Storage of Submissions and Results

The system provides a centralized database for storing student details, assignment submissions, repository links, and evaluation records. This ensures that all academic data is organized and easily accessible whenever required. Faculty members can quickly retrieve submission history and performance details for review. Students can also access their previous records and track their progress over time. Centralized storage improves data management and reduces the risk of information loss.

6.2.4 GitHub Integration for Repository-Based Submissions

An important expected outcome is the successful integration of GitHub as the primary submission platform. Students are encouraged to submit their assignments using GitHub repositories, which promotes the use of version control systems. This helps students gain practical exposure to industry-standard development tools. Faculty members can easily monitor repository activity and access the latest submissions. This integration bridges academic learning with professional software development practices.

6.2.5 Scalable Solution

The proposed system is expected to support a large number of student submissions without affecting performance. As class sizes increase, manual evaluation methods become inefficient and difficult to manage. IntegrityX is designed as a scalable solution capable of handling multiple assignments, repositories, and evaluation records simultaneously. This makes it suitable for deployment across departments and institutions. Scalability ensures long-term usability and adaptability of the system.

6.3 Benefits

6.3.1 Fair and Unbiased Evaluation

The system ensures fairness by following a consistent evaluation workflow for all students. Every assignment is processed uniformly, which removes variations caused by manual grading methods. This leads to more reliable and objective assessment results. Fair evaluation improves student confidence in the academic process and ensures equal opportunities for all learners.

6.3.2 Increased Transparency in Grading

The grading process becomes more transparent as students and faculty can view submission status, evaluation details, and performance records through dashboards. This visibility

helps students understand their progress and encourages them to improve their coding skills. Transparent grading also reduces disputes related to marks and assessment criteria.

6.3.3 Reduced Manual Workload for Faculty

The system significantly reduces the burden on faculty members by automating submission handling and result organization. Instead of manually checking each program, faculty can review consolidated records and performance summaries. This saves time and improves productivity in academic environments.

6.3.4 Better Tracking of Student Performance

The expected outcome includes better monitoring of student progress over time. Faculty members can analyze performance trends, identify weak areas, and provide personalized support. Students can also track their own academic growth, which motivates continuous improvement.

6.3.5 Industry-Level Skill Development

By using GitHub for assignment submissions, students develop practical skills in version control, repository management, and collaborative development workflows. These skills are highly relevant in professional software engineering environments and improve employability.

CHAPTER 7

SYSTEM OUTPUTS AND RESULTS

7.1 Overview

This chapter presents the implementation results and graphical user interface outputs of the IntegrityX system. The developed system was successfully tested for different user roles including students and faculty members. The screenshots below demonstrate the working modules of the system and validate the successful implementation of the proposed methodology. The results highlight the effectiveness of GitHub-based assignment submission and automated evaluation in reducing manual effort and improving consistency in grading. Overall, the outputs confirm that the system provides a structured and user-friendly platform for academic programming assessments.

7.2 User Interface and Implementation Screens

The following figures illustrate the user interface and execution results of the IntegrityX system.

7.2.1 Student Login Page

The student login page provides secure authentication access for students to log in and access their personalized dashboard.

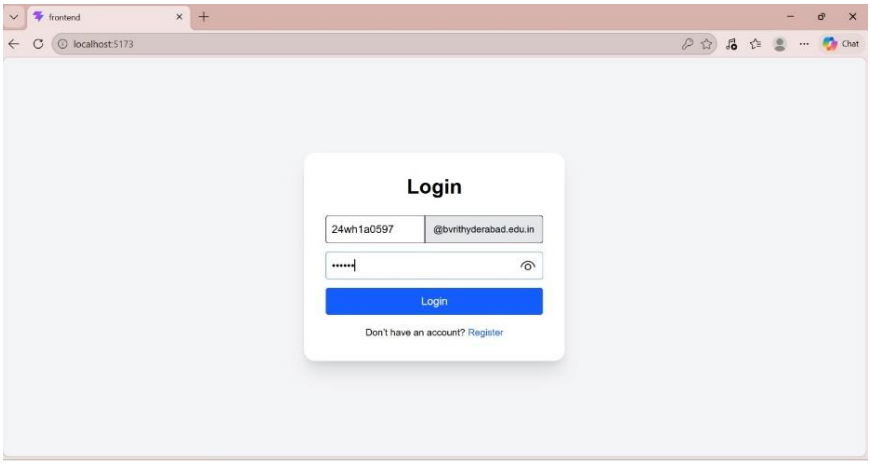


Figure 7.2.1.1: Student Login Interface

7.2.2 Faculty Login Page

The faculty login page enables faculty members to securely access administrative functionalities such as assignment creation, submission review, and marks allocation

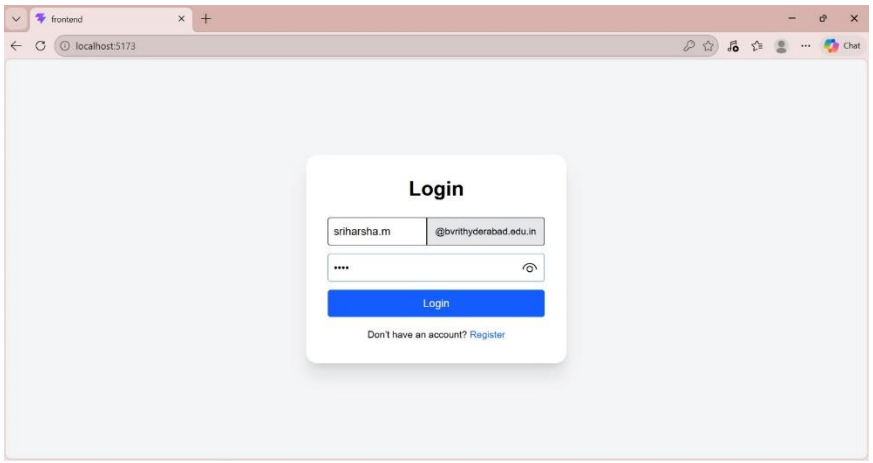


Figure 7.2.2.1: Faculty Login Interface

7.2.3 Student Registration Page

This interface allows new students to register into the system by providing required academic and GitHub-related details.

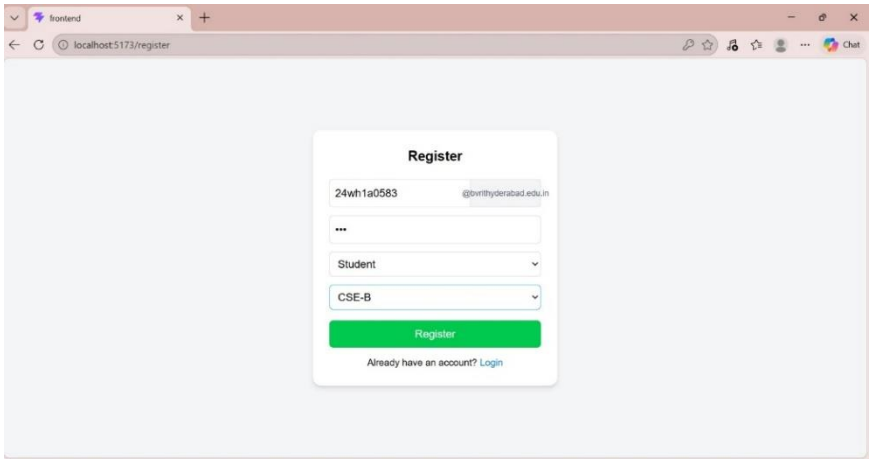


Figure 7.2.3.1: Student Registration Interface

7.2.4 Student Dashboard

The student dashboard acts as the central workspace for students where they can view assignments, submission status, and performance details.

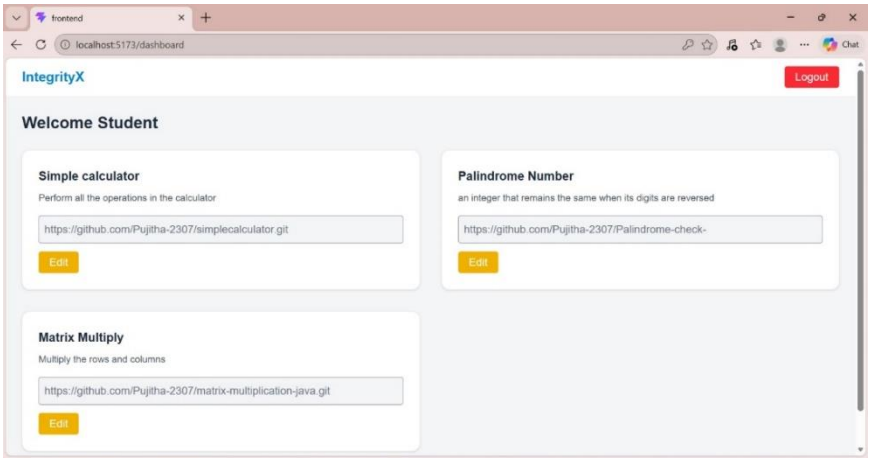


Figure 7.2.4.1: Student Dashboard

7.2.5 Faculty Dashboard – Assignment Creation

This screen shows the faculty interface for creating assignments by specifying assignment title, description, marks, and evaluation parameters.

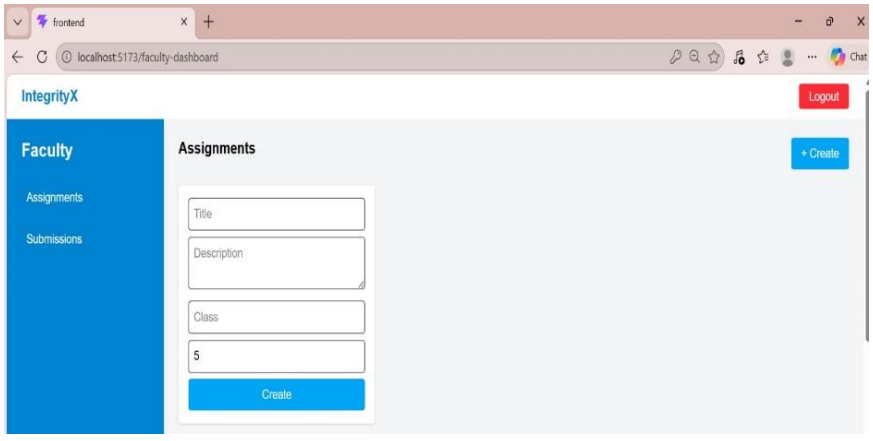


Figure 7.2.5.1: Faculty Dashboard – Assignment Creation

7.2.6 Faculty Dashboard – Assignment View

This interface allows faculty members to view all created assignments and manage existing tasks. It also provides options to monitor assignment details such as edit, delete, descriptions, and evaluation settings for effective administration.

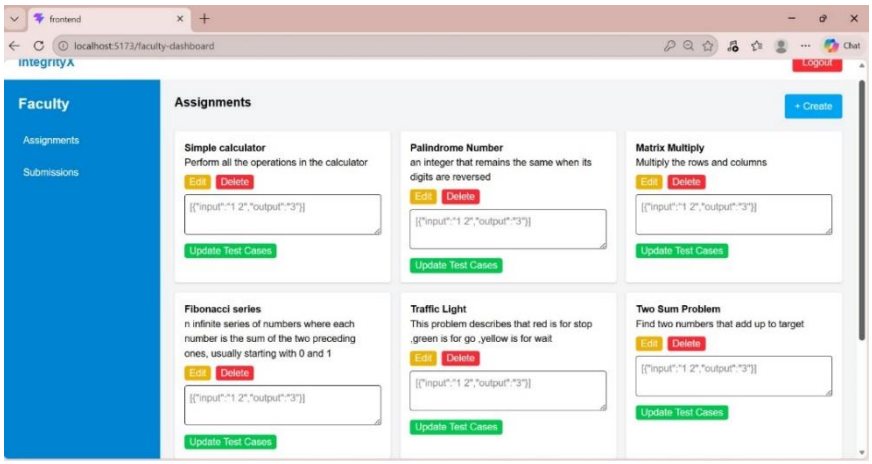


Figure 7.2.6.1: Faculty Dashboard – Assignment View

7.2.7 Faculty Dashboard – Submissions

The submissions panel displays all student repository submissions, enabling faculty members to review code and assign marks.

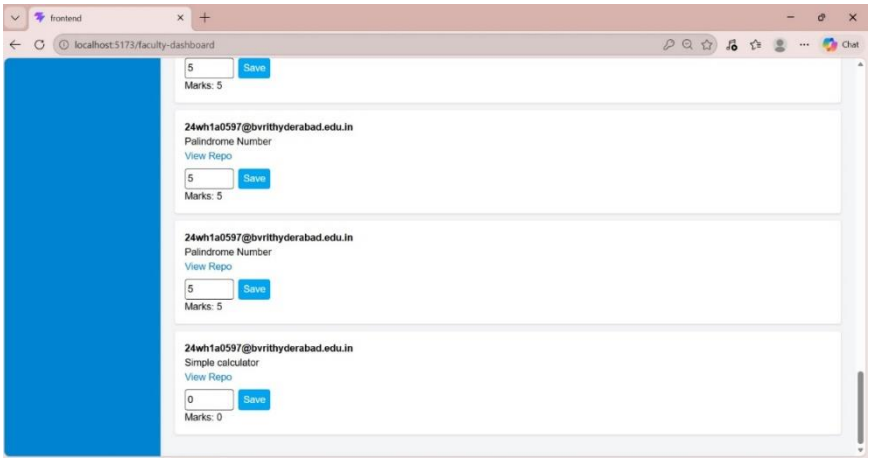


Figure 7.2.7.1: Faculty Dashboard – Submissions

7.3 Result Analysis

The implementation results confirm that the IntegrityX system successfully supports role-based access, assignment management, GitHub repository submission, and faculty-side evaluation. The developed interface ensures ease of use for both students and faculty members, while significantly reducing manual workload and improving transparency in programming assignment assessment. The implementation screens and output interfaces clearly demonstrate the successful working of key functionalities such as secure login, student registration, assignment creation, assignment viewing, submission management, and evaluation workflow. These results validate that the system has been implemented according to the proposed methodology and system architecture discussed in earlier chapters. Furthermore, the integration of GitHub-based submission workflows with automated result management highlights the effectiveness of the solution. Overall, the results confirm that IntegrityX achieves its objective of enhancing efficiency, fairness, transparency, and organization in academic programming assignment evaluation.

CHAPTER 8

8.1 Conclusion

The IntegrityX system effectively addresses the limitations of traditional manual evaluation of programming assignments in academic institutions by reducing the time and effort required from faculty members and improving overall efficiency. By integrating GitHub repositories with an automated backend evaluation mechanism, the system streamlines the complete assessment workflow and enables students to submit assignments through dedicated repositories, encouraging the adoption of version control and industry-standard software development practices. Faculty members benefit from a centralized and well-organized platform where assignments, submissions, and evaluation records are systematically maintained, simplifying academic assessment management. The system also enhances transparency and fairness in grading by ensuring that all submissions are evaluated using predefined and standardized criteria, thereby minimizing human bias and maintaining consistency in results. Additionally, the structured storage of submission history and performance records improves the reliability of assessment and supports scalable evaluation for large student groups. Overall, IntegrityX serves as a modern, scalable, and practical solution that bridges the gap between academic evaluation methods and real-world software engineering workflows, while benefiting both faculty and students through automation, consistency, and professional tool exposure.

8.2 Future Work

Although the current implementation of IntegrityX provides a strong foundation for automated assignment management and evaluation, there is significant scope for further enhancement to make the system more robust and intelligent.

8.2.1 Identification of Missing Submissions

One of the major future enhancements is the ability to automatically identify students who have not submitted their assignments before the specified deadline. The system can be extended to track assignment deadlines and compare them with repository activity. This feature will help faculty members quickly identify non-submissions and take necessary academic actions. It also improves monitoring and ensures better student accountability.

8.2.2 Plagiarism Detection

Another important area of future work is the integration of plagiarism detection mechanisms. The system can be enhanced to compare code submissions across multiple student repositories and identify similarities in logic, structure, or syntax. This will help in maintaining academic integrity and preventing unfair practices during assignment submission.

8.2.3 Advanced Analytics and Reporting

Future versions of the system can include performance analytics modules that generate detailed reports on student progress, ranking, and coding performance trends. Faculty members can use these reports to identify learning gaps and provide targeted support. Students can also benefit from progress tracking dashboards that encourage continuous Improvement

8.2.4 Real-Time Feedback System

Another possible enhancement is the addition of an instant feedback system that provides students with evaluation results immediately after submission. This would enable them to identify errors quickly and resubmit improved solutions within the deadline, thereby enhancing the learning experience.

8.2.5 Scalability for Institutional Deployment

The system can also be expanded for large-scale deployment across multiple departments and institutions. Additional modules for role-based access control, cloud-based storage, and multi-course management can be introduced to support institutional-level academic environments.

References

[1] J. Lee and H. Kim, “Design of Online Judge Systems for Programming Courses,” IEEE Transactions on Learning Technologies, vol. 12, no. 4, pp. 523–534, 2019.

[2] C. Brown et al., “Integrating GitHub Classroom in Software Engineering Education,” ACM Conference on Innovation and Technology in Computer Science Education, pp. 210–215, 2020.